TABLEAU DASHBOARD & VISUALIZATION REPORT

Dashboard architecture, interaction design, KPI quality checks, and presentation standards

Project period: 2010–2024 **Category:** Business Intelligence & Data Analytics

Tools: MySQL, Tableau Public, Microsoft Excel **Source format:** CSV

1. Dashboard Purpose

The Tableau component presents the analysis in a visual, interactive format. The project includes a Home/navigation page and four dashboards organized by business question. The intended visual language uses a dark navy base, blue panels, cyan accents, and light text to create a consistent BMW-inspired interface.

2. Dashboard Architecture

Home / Navigation page

Acts as the entry point and helps users move to the four analytical sections. Confirm that each navigation object opens the correct destination and that the Home page is readable at the intended dashboard size.

Dashboard 1 — Executive Overview

Provides a consolidated, high-level view of the available performance metrics. KPI cards should show clear metric names, appropriate units, and consistent aggregation logic. Trend charts should use a clearly labeled year field and avoid implying causation.

Dashboard 2 — Model Performance

Supports comparisons across BMW models, including ranked model views such as top-five or top-ten lists where implemented. The title and axis should identify the measure being ranked, and filters should make it clear whether the ranking updates dynamically.

Dashboard 3 — Regional Analysis

Explores the distribution of available measures across geographic regions. Regional totals, rankings, and year trends should use consistent units and respond correctly to relevant filters. Avoid labeling a region as best overall unless the metric and period are specified.

Dashboard 4 — Vehicle & Attribute Analysis

Explores the dataset’s vehicle attributes, including available fields such as fuel type, transmission, color, engine size, mileage, and price. Chart choices may include category comparisons, a color distribution donut, fuel-type price

comparison, and a model-by-fuel-type heatmap where these are present in the finished dashboard.

3. Visual Design System

Element	Suggested value	Usage
Main background	#101820 or #050B18	Dashboard canvas and outer frame.
Panel background	#071A3A	KPI cards and chart containers.
Blue accent	#1769AA	Primary emphasis and chart highlights.
Cyan accent	#27C2B8 or #20E3E8	Secondary emphasis, borders, and selected states.
Text	White / light neutral	Titles, labels, and key numbers for contrast.

4. Filter and Interaction Design

- Use concise filter names such as Year, Region, Model, Fuel Type, and Transmission where applicable.
- Keep filter placement and styling consistent across dashboards.
- Test each filter with multiple selections and verify that intended worksheets update.
- Use dashboard actions or navigation buttons only where their destination is clear.
- Ensure text, dropdown controls, and selected values remain legible against dark backgrounds.

5. KPI and Calculation Quality Assurance

- Confirm each KPI’s source field, aggregation, number format, and unit.
- Distinguish SUM, AVG, COUNT, and COUNTD; they represent different concepts.
- Check whether mileage is in kilometers and whether the displayed value is an average or total.
- Validate sales-volume and price calculations against SQL results.
- If a revenue metric is used, document the calculation and verify it fits the source data grain.
- Confirm that filters affect KPI cards and charts consistently.

6. Chart-Specific Review

Ranked bars

Sort in the intended direction, show readable labels, and state the ranked metric. For top-five or top-ten views, confirm whether the filter is dynamic and whether ties are handled as expected.

Donut charts

Use category labels and/or tooltips so slices are interpretable. Avoid relying on color alone; categories should remain understandable if a viewer cannot distinguish every shade.

Heatmaps

Use a clear legend and ensure darker/lighter intensity maps to the intended measure. Check that totals, blanks, and null categories are not mistaken for zero values.

Trend charts

Use chronological ordering and consistent year granularity. Label the measure and make sure any missing years are understood rather than silently interpreted as zero.

7. Dashboard Testing Checklist

- Open the Home page and test every navigation route.
- Review all dashboards at presentation size and confirm no clipped titles or labels.
- Test each filter, reset behavior, and cross-dashboard navigation.
- Compare selected KPI values with the corresponding SQL result.
- Check sorting, units, tooltip content, and legend titles.
- Export the Home page and all four dashboards at consistent resolution.
- Save a packaged workbook (.twbx) where supported so data dependencies travel with the workbook.

8. Export and File Management

Store the Home image and Dashboard 1–4 exports in the 4_Dashboard_Exports folder. Keep the editable Tableau workbook in 3_Tableau. Use stable, descriptive filenames and avoid mixing draft screenshots with final exports. When sharing the project, verify that the packaged workbook opens on another location or machine if possible.

9. Known Limitations

The report describes dashboard purposes and QA practices based on the project structure; it does not certify that every calculation or interaction has been independently tested. A final pass should verify the actual workbook, data source, filter behavior, and displayed values before the dashboard is presented as complete.

10. Conclusion

A professional dashboard is more than a visual design: it must be readable, interactive, numerically accurate, and consistent with the analytical layer. Final quality depends on checking every KPI, filter, ranking, chart label, navigation route, and exported image.